

Semiconductors & Strategic Materials

Initiating Coverage | The Sustainability of the AI Boom

Analyst: Joshua Jameson Date: 08 July 2026 Exchanges: NASDAQ NYSE LSE ASX Euronext Names Covered: 8

INVESTMENT THESIS

The companies that will outperform in this environment are those that are focused on the continuation and sustainability of the AI boom.

MACRO THESIS

The AI boom has driven unprecedented demand across the semiconductor and materials industries - its continuation now depends on addressing critical infrastructure constraints. The shift from training AI models to using them at scale is accelerating energy consumption, memory requirements, and supply chain pressures that the market has not yet fully priced. Energy demands from AI data centres alone are forecast to require an additional 92 gigawatts of power by 2027 (Deloitte/Goldman Sachs, 2026). The winners in this environment will be the companies solving these problems - those providing the next wave of energy, the memory capacity, and the materials supply chain that this next phase of AI infrastructure demands. Government policy is actively accelerating this, with the US CHIPS Act and critical minerals frameworks reshaping how capital is directed across the industry. The sustainability of the AI boom is the investment thesis, and my coverage universe is built around the companies that will determine whether it holds.

The universe is organised around **four structural pillars driving the sector thesis**: Memory shortage, the move from training to inference demand, US industrial policy, and strategic materials.

COVERAGE UNIVERSE

Company	Ticker	Exchange	Pillar	Rating	Rationale
Micron Technology	MU	NASDAQ	Memory Shortage	BUY	Leading HBM manufacturer, set up to take advantage of structural DRAM shortage and accelerating inference demand (HBM demand)
SK Hynix	SKHY	NASDAQ (ADR 10 Jul)	Memory Shortage	BUY	Market leader in HBM supply to multiple leading AI chipmakers; NASDAQ ADR listing this week is an immediate catalyst
TSMC	TSM	NYSE / Euronext	Training to Inference	BUY	Advanced packaging capacity constraint creates pricing power, the indispensable node in AI chip manufacturing, set to benefit from shift towards chiplet infrastructure
ASML	ASML	Euronext Amsterdam	Training to Inference	BUY	Sole global supplier of EUV lithography equipment, no credible competitor exists at the leading edge
Intel	INTC	NASDAQ	US Industrial Policy	BUY	New CEO Lip-Bu Tan turnaround gaining credibility, largest CHIPS Act beneficiary with long-term hyperscaler agreements and US government backing
Cameco	CCJ	NYSE	Strategic Materials	BUY	Largest listed uranium producer, set to benefit directly if uranium becomes next generation energy source for AI infrastructure
Rio Tinto	RIO	LSE	Strategic Materials	BUY	Copper is a significant component in data centre infrastructure, AI infrastructure buildout demand not yet fully reflected in valuation
Lynas Rare Earths	LYC	ASX	Strategic Materials	HOLD	Only non-Chinese rare earth producer and processor, supply agreement with Pentagon validates strategic moat but valuation lofty after 130% rally

**HBM – High Bandwidth Memory, DRAM – Dynamic Random-Access Memory, EUV – Extreme Ultraviolet*

SK HYNIX

TICKER: SKHY (NASDAQ ADR) | RATING: BUY | PILLAR: MEMORY SHORTAGE

As a leading HBM provider, with strong relationships with leading AI chipmakers, SK Hynix is strategically positioned to directly benefit from the structural shortage of DRAM memory, as well as the increasing inference demand. The company is listing on the NASDAQ this Friday (10 July 2026) which will allow it access to large pools of previously inaccessible capital - representing a concrete, time-sensitive catalyst that makes this the most compelling immediate opportunity in the coverage universe.

KEY RISKS

AI capex reversal	Hyperscaler infrastructure spend is partly driven by competitive fear rather than fully monetised workloads. Any material reduction in capex guidance would reprice the entire coverage universe.
China escalation	Further export control tightening could restrict H2O sales and DUV equipment exports, compressing revenue for ASML and disrupting supply chains dependent on Chinese processing capacity.
Memory overbuild	Despite current discipline, a rapid capacity expansion response to high HBM pricing could recreate historical boom-bust dynamics for Micron and SK Hynix.
Uranium timeline risk	Nuclear PPA commitments are long-dated. If permitting delays or cost overruns slow reactor restarts, the uranium demand catalyst for Cameco is deferred, not eliminated.
Lynas execution & valuation	CEO transition and a 226x earnings multiple leave limited margin of safety. Any operational setback at the Texas or Kalgoorlie facilities would compress the valuation sharply.